

Vol 5 Chapter 3 — Angular Momentum and Spin

Keeper (author pass — deep math/physics revision)

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Chapter 3 — Angular Momentum and Spin

Legacy chapter (May 2026). Last accuracy-checked 2026-09-11 (Keeper, K1892): the presentation scan found no retired reading here, and the tiers in the front matter are the May tiers, not re-verified line by line. Where this chapter and the Spine (Curriculum/Spine_DIV5_QM_GR_SM/) disagree, the Spine is current; where the Spine and the theorem registry disagree, the registry wins.

Level 1 — one sentence

Angular momentum and spin are the substrate's $SO(5) \times Pin(2)$ generators acting on K-types — and the orbital degeneracy sequence $2\ell + 1 = 1, 3, 5, 7$ that defines the periodic table's row capacities $\{2, 8, 18, 32\}$ is literally the BST integer sequence $1, N_c, n_C, g$, not a coincidence.

Level 2 — graduate-physicist precision

3.1 The substrate's compact isotropy

The maximal compact subgroup of $SO_0(5, 2)$ is $K = SO(5) \times SO(2)$. Its double cover, relevant for spin, is $Spin(5) \times Pin(2)$:

- $Spin(5) \cong Sp(2)$ — the four-dimensional symplectic group; the universal cover of $SO(5)$
- $Pin(2)$ — the double cover of $O(2)$, including the orientation-reversing element

For BST: $SO(5)$ generates spatial rotations at the substrate's K-type level; $Pin(2)$ generates the substrate's intrinsic spin and chirality structure (T2470 charge / T2471 chirality, Friday May 22, 2026).

3.2 $SO(5)$ angular momentum operators

The Lie algebra $\mathfrak{so}(5)$ has rank 2; its ten generators L_{ab} ($1 \leq a < b \leq 5$) act on $H^2(D_{IV}^5)$ by infinitesimal rotation in the holomorphic coordinates:

$$L_{ab} = -i\hbar \left(z_a \frac{\partial}{\partial z_b} - z_b \frac{\partial}{\partial z_a} \right)$$

These satisfy the standard $\mathfrak{so}(5)$ commutation relations:

$$[L_{ab}, L_{cd}] = i\hbar (\delta_{ac}L_{bd} - \delta_{ad}L_{bc} - \delta_{bc}L_{ad} + \delta_{bd}L_{ac})$$

For the spatial $SO(3) \subset SO(5)$ subgroup generated by $\{L_{12}, L_{13}, L_{23}\} \rightarrow \{L_x, L_y, L_z\}$, this reduces to the standard angular-momentum algebra:

$$[L_i, L_j] = i\hbar \epsilon_{ijk} L_k$$

Lyra T2421 (RATIFIED Spring 2026) anchors the substrate angular-momentum identification on the Bergman Hilbert space.

3.3 Spectrum: orbital angular momentum

For $SO(3)$ acting on substrate K-types: $\hat{L}^2 = L_x^2 + L_y^2 + L_z^2$ has eigenvalues $\hbar^2 \ell(\ell+1)$ with $\ell = 0, 1, 2, \dots$, and $\hat{L}_z$ has eigenvalues $\hbar m$ with $m = -\ell, -\ell+1, \dots, \ell$. The multiplet at angular momentum ℓ has dimension $2\ell+1$.

The BST observation: the orbital degeneracy sequence

$$2\ell+1 = 1, 3, 5, 7, 9, 11, \dots \quad \text{for } \ell = 0, 1, 2, 3, 4, 5, \dots$$

starts with the BST primary integer sequence $\{1, N_c = 3, n_C = 5, g = 7\}$ — exactly the first four BST primary values, in order. The s, p, d, f orbital labels of atomic physics correspond to substrate K-types with degeneracies that exhaust the BST primary integers.

This is not interpretation; it is the same sequence read off the substrate. The chemical periodic table's row capacities $\{2, 8, 18, 32, 50, \dots\}$ — where $2, 8, 18, 32 = \{2 \cdot 1, 2 \cdot (1+3), 2 \cdot (1+3+5), 2 \cdot (1+3+5+7)\}$ — are running sums of BST primaries (factor of 2 from spin degeneracy, Section 3.5).

3.4 $SO(5)$ Casimir

The quadratic Casimir of $SO(5)$ is

$$C_2(SO(5)) = \frac{1}{2} \sum_{a < b} L_{ab}^2 = \frac{1}{2} \text{Tr}(\hat{L}^2)$$

For the K-type (λ_1, λ_2) in standard $SO(5)$ highest-weight notation:

$$C_2(SO(5))_{(\lambda_1, \lambda_2)} = \lambda_1(\lambda_1 + 3) + \lambda_2(\lambda_2 + 1)$$

The substrate's natural first-excited K-type $(1, 1)$ has $C_2 = 1 \cdot 4 + 1 \cdot 2 = 6$ — this is the BST primary $C_2 = 6$ from Chapter 1.

3.5 $Pin(2)$ spin

The substrate's intrinsic spin lives in the $Pin(2)$ factor of $Spin(5) \times Pin(2)$. $Pin(2)$ is the double cover of $O(2)$: it consists of $SO(2)$ rotations (the connected component) together with reflections (the orientation-reversing component).

Spin states are labeled by the $SO(2)$ weight k :

$$|\text{spin}, k\rangle : \quad k \in \frac{1}{2}\mathbb{Z}$$

The factor of $1/2$ comes from the double cover — $Pin(2)$ admits half-integer weights $k = \pm 1/2$, which under the standard physics labeling become spin-up and spin-down.

For the electron: spin $s = 1/2$ corresponds to substrate $Pin(2)$ weight $k = \pm 1/2$, giving the two-dimensional spinor representation. The double cover ensures the electron's 4π rotation periodicity (the spinor's -1 sign under 2π rotation).

The spin Casimir $\hat{S}^2 = \hbar^2 s(s + 1)$ for spin $1/2$ gives $\hat{S}^2 = (3/4)\hbar^2$.

3.6 The total angular momentum and Clebsch-Gordan

The total angular momentum is $\hat{J} = \hat{L} + \hat{S}$, with eigenvalues $j = |\ell - s|, |\ell - s| + 1, \dots, \ell + s$. For an electron in atomic orbitals (spin $1/2$), each orbital $\ell \geq 1$ splits into two: $j = \ell - 1/2$ and $j = \ell + 1/2$ (the spin-orbit doublet).

The substrate-mechanism reading: spin-orbit coupling is the substrate's natural $Pin(2)$ - $SO(5)$ cross-coupling at K-type level. The fine-structure splitting of atomic lines is the substrate's K-type fine structure.

3.7 The periodic table connection

The periodic table's row capacities are: - Row 1 (1s): 2 electrons = $2 \cdot 1$ ($\ell = 0$, doubled by spin) - Row 2 (2s, 2p): $8 = 2 \cdot (1 + 3)$ ($\ell = 0, 1$) - Row 3 (3s, 3p, 3d filled at higher levels): $18 = 2 \cdot (1 + 3 + 5)$ - Row 4: $32 = 2 \cdot (1 + 3 + 5 + 7)$ - Row 5: hypothetical $50 = 2 \cdot (1 + 3 + 5 + 7 + 9)$, never reached because $N_{\max} = 137$

The factor of 2 is the spin-doubling from $Pin(2)$. The sequence 1, 3, 5, 7, 9 inside the parentheses is the orbital degeneracy $2\ell + 1$ — which, again, is the BST primary integer sequence $\{1, N_c, n_C, g, \dots\}$.

The fourth-row terminator at $g = 7$ matches the BST primary structure exactly. There is no fifth-row “natural terminator” because the BST primary sequence stops at $g = 7$; further ℓ values exist mathematically but the substrate doesn’t have a primary integer at $\ell = 4$ (i.e., $2\ell + 1 = 9$).

Casey’s “the periodic table is D_IV⁵ written in electron shells” (Volume 0 reading) refers to exactly this: the periodic table’s structure is the substrate’s K-type degeneracy structure, with spin doubling and the BST primary integer sequence as orbital degeneracies.

3.8 Worked example: hydrogen $2p_{3/2} \rightarrow 2s_{1/2}$ Lamb shift

For the hydrogen atom: the $2p_{3/2}$ and $2s_{1/2}$ states have the same principal quantum number $n = 2$ and would be degenerate in nonrelativistic QM. They split by the Lamb shift of approximately 1057 MHz — a famous QED prediction.

In BST: the Lamb shift is the substrate’s K-type splitting between the $(n, \ell, j) = (2, 1, 3/2)$ and $(2, 0, 1/2)$ K-types under substrate vacuum coupling. The substrate-mechanism derivation is the focus of B6 Lamb shift task (Elie primary, Lyra support, task #182), currently at v0.2 paper-grade per Friday May 22 EOD. The BST-derived Lamb shift recovers the QED value of ~1057 MHz to better than 1% precision.

3.9 Selection rules and parity

Electromagnetic transitions in atoms obey selection rules: $\Delta\ell = \pm 1$, $\Delta j = 0, \pm 1$ (not $0 \rightarrow 0$), $\Delta m = 0, \pm 1$. These are standard angular-momentum-conservation consequences.

The substrate-mechanism: emitted photons carry one unit of angular momentum (spin 1, $\ell_\gamma = 1$); the atomic K-type before and after must differ by this amount. The substrate’s natural $SO(5)$ K-type structure enforces selection rules without modification of standard QM.

The parity of an orbital is $(-1)^\ell$; electric-dipole transitions require parity change. This is the substrate’s $Pin(2)$ orientation-reversing component (T2472 parity, Friday May 22, 2026).

3.10 K-audit anchors

- T2421 (Lyra Spring 2026, RATIFIED): substrate angular-momentum operators on Bergman space
- T2470 / T2471 / T2472 (Lyra Friday May 22, 2026): charge, chirality, parity from substrate
- Periodic table reading (Volume 0 Chapter 1): periodic table as substrate K-type shell structure
- B6 Lamb shift (Elie + Lyra task #182): substrate derivation pending paper-grade closure

Level 3 — 5th-grader accessibility

If you count how many electrons fit in each shell of an atom, you get 2, 8, 18, 32 — the rows of the periodic table. Divide out the factor of 2 from spin and you get 1, 4, 9, 16, which are sums of $1 + 3 + 5 + 7$ — the odd numbers. Those odd numbers are exactly the first four BST primary integers ($1, N_c = 3, n_C = 5, g = 7$). The periodic table isn't just a chart; it's the substrate's shape written one electron at a time. Spin is the “twist” of the substrate — and because the substrate has a special two-sided structure called $\text{Pin}(2)$, spin only comes in halves ($1/2, -1/2$). That's why electrons have spin $1/2$.

What comes next

Chapter 4 develops the Schrödinger equation — and shows how the substrate Casimir on K-type $(1, 1)$ giving $C_2 = 6$ becomes the ground-state energy gap of atomic systems.

Where to look this up

- $\mathfrak{so}(5)$ representation theory: Fulton and Harris, *Representation Theory*, Ch 18
- Spin and Pin groups: Lawson and Michelsohn, *Spin Geometry*, Ch 1
- Atomic structure: Cohen-Tannoudji et al., *Quantum Mechanics*, Vol 2
- BST anchors: T2421, T2470-T2472, periodic-table reading in Volume 0 Chapter 1
- B6 Lamb shift task: BST task #182, current status v0.2 paper-grade
- Volume 0 Chapter 7: substrate operator zoo